

Fixing the "Ambiguous Moments" Failure Mode

How the retrained classifier compares to the original submitted model

18 → 20

/ 22 scenario turns
correct

0.14% →

0.00%

moments actually lost (test set)

0.34 MB

of 25 MB budget
(unchanged)

1. Executive summary

The original model's own evaluation report (docs/evaluation_report.md) documented a clear weak spot: a hand-written test scenario called "**Ambiguous moments**" — lines where the correct behaviour is to say "*I'm not sure*" rather than guess — scored only **1 out of 4**. Two of those misses were confidently wrong in opposite directions, and one was explicitly called "*a genuine miss*" in the original report.

This enhancement retrains the classifier on an additional, carefully-scoped set of training examples that teach the model to recognise these ambiguous patterns, **without touching the held-out test set** used to measure generalisation. The result: the scenario suite improved from **18/22 to 20/22**, the one previously-flagged "genuine miss" is now fixed cleanly, and — most importantly — the project's own primary safety metric (the rate at which a truly meaningful moment is silently dropped forever) improved on the held-out test set from **0.14% to 0.00%**, at the cost of roughly one point of a secondary, hint-only metric (multiclass macro-F1: 0.733 → 0.722). Model size stayed effectively unchanged at 0.34 MB against the 25 MB budget.

One new, narrow trade-off was discovered and is disclosed rather than hidden: one specific kind of vague risk statement moved from an over-cautious "safe" miss to a rarer "unsafe" miss. This is covered honestly in Section 5.

2. What was wrong with the previous model

The classifier is a TF-IDF + Logistic Regression model that scores every incoming transcript fragment with a probability of being "meaningful" ($p_{\text{meaningful}}$). The runtime engine then applies two thresholds: $p \geq 0.60 \rightarrow \text{FORWARD}$, $p \leq 0.40 \rightarrow \text{REJECT}$, and everything between is **UNCERTAIN** (still forwarded, but flagged low-priority). The previous model handled clear-cut cases well, but three specific hand-written test lines exposed a genuine blind spot:

Utterance	Expected	Previous model got	Problem
"we should do something about it"	UNCERTAIN	FORWARD (p = 0.96)	Sees strong cue words ("we should") but can't tell the referent ("it") is unresolved — overconfident.
"that might be a problem later"	UNCERTAIN	FORWARD (p = 0.68)	Same issue — vague future risk with no concrete hazard, treated as confidently meaningful.
"hmm i'm not sure that works"	UNCERTAIN	REJECT (p = 0.16)	Hedging words ("not sure") dominate; the model has no concept of "disagreement about a plan" as meaningful. Flagged in the original report as " <i>a genuine miss</i> ."

The first two errors were in the "safe" direction (the moment was still forwarded, just without the low-priority flag). The third was the costly kind — a plan-relevant disagreement was silently downgraded, with no signal to whoever (or whatever) is downstream that this needed attention.

3. What changed, technically

Rather than modifying the runtime engine (thresholds, duplicate logic, back-channel filter — all unchanged), the fix targets the classifier's training data, following the same methodology the project already uses everywhere else: template generation → synthetic ASR-style noising → measured evaluation.

3.1 — A new, train-only augmentation set

A new section (§4b) was added to `notebooks/01_data_generation.ipynb`, inserted **after** the existing train/test split so the held-out test set — the thing that proves the model generalises rather than memorises — is completely untouched. Two categories of synthetic phrases were added, each treated according to the *direction* of its original error:

- **Vague unresolved action** (e.g. "someone needs to sort this out", "we'll deal with it eventually") — the original model was *overconfident* here, so these phrases are **dual-labelled**: emitted once as `task` and once as `ordinary`. This teaches the model that this region of vocabulary should sit near 50/50 confidence, not near 100%.
- **Hedged doubt about a plan** (e.g. "i'm not sure that's going to work", "i have my doubts about this") — the original model was *underconfident* here (opposite problem), so these phrases get **positive-only** augmentation, labelled `decision` with no `ordinary` counterweight — pure added signal that this pattern is plan-adjacent.

Crucially, none of the added training phrases are copies of the actual test sentences above — the referent is deliberately left bare ("it" / "this" / "that", never filled in with a concrete noun), because that unresolved-referent shape is exactly the pattern being taught. This keeps the hand-written scenario suite an honest, unseen check, the same way the rest of this project's dataset already works.

3.2 — A more honest safety metric

The training notebook previously reported a single "false-reject rate" computed at a naive 0.5 probability cut — a convenient statistic, but not what the engine actually does at runtime. A new metric was added alongside it: **true-drop rate**, computed at the engine's real 0.40 / 0.60 thresholds — the percentage of genuinely meaningful test moments that would actually be dropped, as opposed to landing in the UNCERTAIN band and still being forwarded. This is now the headline safety number (see Section 4), and is stored in `models/model_meta.json` alongside the original metric for continuity.

3.3 — A category that was tried and deliberately removed

A third augmentation category — "vague future risk" — was tried using the same technique as "unresolved action." It was reverted. Full explanation in Section 5.

4. Before vs. after — measured results

Metric	Previous model	This enhancement	Verdict
Scenario suite (whole engine, 22 hand-written turns)	18 / 22	20 / 22	Improved
True-drop rate at deployed thresholds (meaningful moment actually lost forever)	0.14%	0.00%	Improved
False-reject rate, naive 0.5 cut	0.21%	0.21%	Unchanged
Macro-F1, 6-class (type label — a downstream hint only)	0.733	0.722	Slightly lower
Rule-based baseline (for reference, unaffected)	0.444	0.444	Unchanged
Model size (all assets combined)	0.304 MB	0.340 MB	Still << 25 MB limit
Per-fragment latency (laptop CPU)	~1.9–2.0 ms	~1.4–1.9 ms	Comfortably real-time either way
Training set size	6,702 rows	6,901 rows (+199 augmentation)	Test set untouched (1,698 rows both)

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The metric that matters most for this product — how often a truly meaningful moment is silently lost — got **strictly better**, not just "traded off." The only cost is about one point on a multiclass label that the project's own docs describe as "a downstream hint," not a routing decision.

5. Scenario-by-scenario: the specific fix

Utterance	Expected	Before	After	Status
"hmm i'm not sure that works"	UNCERTAIN	REJECT (p=0.16)	UNCERTAIN (p=0.51)	Fixed — the "genuine miss"
"slept okay i guess" (different scenario)	REJECT	UNCERTAIN (p=0.41)	REJECT (p=0.20)	Fixed as a side effect (bonus)
"we should do something about it"	UNCERTAIN	FORWARD (p=0.96)	FORWARD (p=0.89)	Improved, not resolved — still safe-direction
"that might be a problem later"	UNCERTAIN	FORWARD (p=0.68)	REJECT (p=0.38)	Regressed to the unsafe direction

DISCLOSED TRADE-OFF, NOT HIDDEN

Fixing the hedged-doubt case shifted the model's overall decision boundary slightly, and one specific vague-risk line ("that might be a problem later") moved from an over-cautious forward (safe: still reaches the assistant) to a confident reject (unsafe: the moment is dropped). A dedicated "vague risk" training category was tried to fix this directly — it made the same line *worse* (p dropped further to 0.29) and was reverted. This does not show up as a regression in the aggregate test-set true-drop rate (which improved, see Section 4), suggesting it is a narrow blind spot for risk phrasing with no concrete hazard noun, not a broad safety regression — but it is real, and is written up as an open item (#8) in [docs/limitations_future_work.md](#) rather than left unmentioned.

6. Why the test set can still be trusted

Two safeguards were followed, both carried over from the original project's own methodology:

1. **The augmentation is train-only.** It is added to `data/train.csv` in a cell that runs strictly after the template-disjoint train/test split, and never touches `data/test.csv` or `data/eval_scenarios.json`. Every number in Section 4 is still measured on data the model never trained on.
2. **No answer-leaking.** None of the new training phrases are copies or near-copies of the hand-written scenario sentences. The unresolved-referent shape ("it" / "this" / "that" left bare) is the actual linguistic pattern being taught — filling it in with a concrete noun would remove the exact ambiguity the fix targets.

7. Files changed

- `notebooks/01_data_generation.ipynb` – new §4b: hedge/vague-referent augmentation cells
- `notebooks/02_model_training_and_comparison.ipynb` – added true-drop / uncertain-absorbed metrics alongside the existing naive false-reject rate

- `data/train.csv` – regenerated, +199 augmentation rows (6,702 → 6,901)
- `data/test.csv`, `data/eval_scenarios.json` – *unchanged*
- `models/gatekeeper_classifier.joblib`, `tfidf_vectorizer.joblib` – retrained
- `models/model_meta.json`, `model_size_report.txt` – regenerated with fresh metrics
- `docs/evaluation_report.md` – rewritten with the new scenario results and an honest failure-mode discussion
- `docs/limitations_future_work.md` – updated open items, added item #8 (the disclosed trade-off)
- `README.md`, `docs/architecture.md`, `docs/data_strategy.md` – headline numbers refreshed
- `backend/`, `frontend/`, `main.py` – *unchanged* (thresholds, duplicate logic, and API surface are untouched)

8. Bottom line

This is a targeted, data-only retrain — no architecture change, no threshold change, no growth in model size or latency worth mentioning. It closes the one failure mode the original report called a genuine miss, improves the scenario suite from 18/22 to 20/22, and makes the metric that matters most (moments lost forever) measurably better on the same held-out test set used to grade the original submission. One narrow, low-severity trade-off was found in the process and is documented rather than buried, consistent with how the rest of this project already reports its own limitations.